

Task 9: Problem-Solving & Design Explanation

A. Problem decomposition

To program this HomeSense, we broke down various components of the program to smaller sections like checking range, whether the given range for a device is normal, the energy status, cost, and for abnormal cases like a low value or an unknown device or a zero value.

B. Decision making

The reading is considered normal if it is in the given specified range. A reading above the normal range up to 50% is considered high and a value over 50% of the maximum normal range is where it is considered critical. All devices have their own individual normal, high and critical values.

C. Boundary cases

A reading exactly equal to the maximum value in the normal range is considered normal and a reading exactly equal to the minimum value in the normal range is also considered normal. Anything above or below the normal range is not considered normal.

D. Different devices

Different devices have different energy requirements. Some consume more energy whereas others consume less energy. No two devices can have the same normal range. For example, Normal range for LED Light is 0.01-0.10 kWh, Television is 0.05-0.50 kWh, Refrigerator is 0.10-1.50 kWh, Washing Machine is 0.30-2.50 kWh, and Air Conditioner is 0.50-5.00 kWh.

E. Multiple Readings

To analyze multiple readings we used nested lists, which is many lists inside one main list. The list contains the name of the device and its energy consumption. Each of the readings in the list are looped to check their range, rate and status to check if it requires attention. For each of these readings, the range, cost, status and feedback are calculated.

F. Testing

We used the assert function to verify the outputs of our functions.

`test_energy_status()` is used to check if the reading is normal for a normal value

`test_energy_status_boundary()` is used to check if the value is normal for a boundary value.

`test_energy_status_high()` is used to check if the given value is higher than the normal range.

`test_energy_status_critical()` is used to check if the value is critical.

test_energy_status_invalid() is used to check if the given value is low or not valid.

test_energy_cost() is used to check the cost of the energy.

energy_consumption * electricity_rate.

test_attention_normal() is used to see if the value is normal so no attention is required.

test_attention_high() is used to check if the value is high and attention is required.

test_unknown_device() is used to check if a device is known or unknown.

Out of these test functions, the ones that were important to verify the solutions were test_energy_status_high() and test_energy_status_critical() to check if a device is consuming an alarming amount of energy that requires attention. And test_energy_cost() to verify that the correct amount is being charged for the energy consumption.